Trust-Guided Multimodal LLM Integration with Reinforcement Learning for Autonomous Driving

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Abstract

Large language models (LLMs) offer semantic reasoning capabilities valuable for autonomous driving, but their output reliability remains uncertain and scenario-dependent. This work investigates integrating multimodal LLMs with reinforcement learning by introducing a learned trust-gating mechanism that estimates LLM output reliability from sensor state. The proposed approach combines a three-stage LLM reasoning pipeline (perception → planning → control) with transformer-based multimodal sensor fusion. A comprehensive ablation study across 2,250 episodes (15 configurations, 3 random seeds, 10 diverse scenarios) demonstrates that LLM-guided reward shaping achieves 29.3% performance improvement and 47.8% collision reduction compared to baseline RL. Trust floor sensitivity analysis justifies $\tau_{\min} = 0.3$ *as optimal, while fallback quantification reveals that rule-based systems handle 45% of decisions in perception-degraded scenarios. Reward attribution analysis confirms LLM alignment contributes 11.8% improvement independent of trust gating. Analysis reveals LLMs provide substantial value for high-level semantic reasoning (scene understanding: +46% for pedestrian crossing) but limited benefit for routine control tasks (+10% for highway cruise). The learned trust-gating mechanism reduces output variance by 74%, indicating that principled confidence estimation enables safer integration of external semantic guidance in safety-critical control.*

Keywords: *Autonomous Driving; Multimodal LLMs; Reinforcement Learning; Trust-Guided Mechanism; Semantic Reasoning; Safety-Critical Control; Multimodal Sensor Fusion*

1. Introduction

Autonomous driving requires integrating real-time perception, high-level reasoning, and precise control across diverse driving conditions [1, 2]. Real-world autonomous systems face a fundamental reliability challenge: ensuring safety across unbounded scenarios while leveraging emerging semantic reasoning capabilities. Large language models have demonstrated strong capabilities in commonsense reasoning, planning, and semantic understanding [3], opening new possibilities for handling long-tail scenarios and complex driving situations. However, directly applying LLMs to autonomous driving faces distinct challenges: output reliability varies significantly by scenario (ranging from 95% accuracy in clear weather to <50% in adverse conditions), inference latency (50-100 ms) conflicts with real-time control requirements (10 ms cycle), and hallucinations can propagate to safety-critical decisions [4].

Rather than replacing reinforcement learning (RL) policies with LLM outputs directly, this work proposes a complementary approach: using LLMs to generate high-level semantic guidance while training an RL agent to learn *when to trust* this guidance. This strategy preserves RL's formal safety properties while enabling semantic reasoning through a learned confidence mechanism. The key insight: reliable guidance integration requires estimating guidance reliability itself, not assuming static reliability across all scenarios.

Key Contributions of the proposed work:

1. **Learned Trust-Gating Mechanism:** A differentiable network jointly optimized with RL to estimate LLM reliability ($\tau \in [0.3, 1.0]$) from multimodal sensor state, enabling scenario-dependent confidence estimation
2. **Three-Stage Reasoning Pipeline:** Decomposed LLM reasoning (perception → planning → control) with fallback mechanisms grounded in multimodal sensor fusion, respecting the boundary between what (semantic) and how (control)
3. **Comprehensive Experimental Validation:** Systematic ablation across 2,250 episodes (15 configurations, 3

seeds, 50 episodes each), 10 driving scenarios, revealing clear patterns of when LLM guidance helps versus hurts, with 74% variance reduction
4. **Practical Safety Insights:** Demonstrated that learned trust estimation outperforms naive baseline (LLM-only) and provides principled alternative to heuristic thresholding

2. Related Work

Prior work spans end-to-end learning for autonomous driving, LLM applications in robotics, multimodal integration, and uncertainty quantification in RL. We position our approach as combining learned trust estimation with modular architecture design.

End-to-end learning maps raw sensor inputs directly to vehicle control, establishing a foundation for learning-based driving systems [1]. Conditional imitation learning [5] extended this by incorporating high-level navigation commands to improve policy generalization across routes. Recent transformer-based approaches [6] employ multimodal fusion of camera and LiDAR data, achieving improved scene understanding. However, two critical research gaps persist: First, pure RL approaches struggle with out-of-distribution scenarios where semantic reasoning is essential. Second, direct LLM control lacks the frequency and reliability required for safety-critical driving. This work addresses these gaps by integrating explicit semantic guidance through LLMs, maintaining the benefits of learned control while adding reasoning about novel situations.

2.1. Large Language Models for Sequential Decision-Making

LLMs have been explored as task planners for robotic systems [7, 8], demonstrating capability for zero-shot policy generation through prompting. Vision-language models (VLMs) such as LLaVA [9] enable semantic scene understanding from visual data. A key limitation in prior work: direct application of LLM outputs for control fails due to incompatible latency (50-100 ms per inference vs. 10 ms control cycle requirement) and unreliable predictions (especially in adverse conditions). Recent work shows LLM outputs can achieve 80-90% accuracy on semantic tasks but are unstable across scenarios [3, 10]. The proposed approach addresses this through learned confidence estimation, enabling more robust integration of LLM guidance into real-time control loops with explicit reliability assessment.

2.2. Multimodal LLMs for Autonomous Driving

OpenEMMA [11] proposes end-to-end multimodal models jointly learning perception and control. SenseRAG [12] constructs knowledge bases with proactive LLM querying. ScVLM [13] enhances VLMs for safety-critical event perception. Research gap: prior approaches either replace RL with end-to-end models (losing safety properties and interpretability) or focus on perception alone (missing control integration and uncertainty handling). This work uniquely combines modular design with learned trust estimation for principled guidance integration, maintaining the benefits of each component while enabling their safe coordination.

2.3. Uncertainty and Trust in Reinforcement Learning

Bayesian RL and ensemble methods estimate epistemic uncertainty [14], but typically only model observation uncertainty from the environment, not reliability of external guidance. Recent work on LLM uncertainty estimation [15] provides complementary techniques for assessing model confidence. The research gap addressed here: existing methods lack scenario-dependent confidence estimation specifically for LLM guidance in safety-critical systems. Unlike generic uncertainty quantification (aleatoric vs. epistemic), the trust-gating mechanism learns *when to trust* external guidance, which is fundamentally different from estimating observation uncertainty. This enables safe integration while maintaining RL's learned safety properties.

3. System Architecture

The integrated system combines multimodal sensor fusion, three-stage LLM reasoning, and learned trust-gating. The architecture respects the principle that LLMs excel at high-level reasoning (what should happen) while learned control policies excel at low-level execution (how to achieve it).

3.1. High-Level System Design

Figure 1 illustrates the complete system architecture. Three main components integrate seamlessly: (1) multimodal sensor fusion grounds the system with real-world perception; (2) a three-stage LLM reasoning pipeline translates scene understanding to control actions; and (3) a learned trust-gating mechanism decides when to use LLM guidance.

The architecture consists of three tightly integrated components that operate sequentially:

1. **Multimodal Sensor Fusion:** Encodes diverse sensor modalities (camera, LiDAR, IMU, GPS) into a unified 64-dimensional representation via transformer-based cross-modal attention. This provides the low-level perception grounding for both LLM guidance and RL policy.
2. **Three-Stage LLM Reasoning Pipeline:**
   - **Perception Stage (LLaVA):** Camera image → semantic scene description and object detections (high-level scene understanding)
   - **Planning Stage (Phi-3):** Semantic scene → risk assessment and high-level action recommendation (reasoning about what should happen)

- **Control Stage (Phi-3):** High-level action $\rightarrow$ normalized continuous control output $[\theta, \beta, \sigma]$ (translation to actuation)

3. **Trust-Gating Module:** Learned network estimating LLM output reliability from sensor state and LLM outputs, outputting confidence score $\tau \in [0.3, 1.0]$. This module learns scenario-dependent when-to-trust decisions.

3.2. Multimodal Sensor Fusion

Four modalities (camera, LiDAR, IMU, GPS) are independently encoded, concatenated, and processed by a transformer with cross-modal attention to capture multi-sensor dependencies. The final fused 64D representation serves as grounding for both LLM conditioning and trust estimation:

Camera Encoding: ResNet blocks with GroupNorm (for batch-size-1 robustness in online deployment):

Listing 1. Camera encoder: processes RGB image to feature vector

```
# ResNet-style blocks: Conv2d(3,32) ->
    Conv2d(32,64) -> Conv2d(64,128)
# GroupNorm(8,channels) for stable single-
    sample normalization
# AdaptiveAvgPool2d((1,1)) produces 256D
    features
```

LiDAR Encoding: PointNet-style architecture processing point clouds (500 points, 4D: x,y,z,intensity):

Listing 2. LiDAR feature extraction: point cloud to embedding

```
# Three conv layers: (N,4) -> (N,64) -> (N
    ,128) -> (N,256)
# Max pooling across points: (1,256)
# Fully connected layer produces 64D
    features
```

IMU and GPS: Direct embedding layers producing 16D and 8D features respectively. IMU captures acceleration and angular velocity; GPS provides absolute position information.

Cross-Modal Fusion: All modalities projected to 256D, concatenated as sequence (4 tokens), processed through 2-layer transformer (4 attention heads, 512D hidden) following the standard attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Mean pooling over the 4-token sequence produces final 64D multimodal representation. This 64D vector captures the joint information from all modalities and serves as the sensor state for both LLM grounding and trust estimation.

3.3. Three-Stage LLM Reasoning Pipeline

Autonomous driving decomposes into three stages: perception (understanding the scene), planning (reasoning about actions), and control (executing decisions). This design leverages LLM strengths in semantic reasoning while delegating low-level control to learned policies.

3.3.1. Perception Stage

LLaVA-1.5-7B model processes camera images to extract semantic understanding. This stage extracts what the scene contains:

$$P_{\text{llm}} = \text{LLaVA}(I_{\text{camera}}) \rightarrow \{\text{objects, attributes, relationships}\} \tag{2}$$

Output includes object class (car, pedestrian, traffic light, etc.), position (left/center/right, near/far), confidence. JSON parsing with regex extraction attempts structured extraction; fallback uses rule-based computer vision (Canny edge detection for objects, HSV color space for traffic signals). The fallback mechanism ensures robustness when LLM fails to produce structured output.

3.3.2. Planning Stage

Phi-3 mini evaluates scenario risk and proposes high-level actions. This stage performs reasoning about what should happen given the scene:

$$\mathcal{P} = \text{Phi3}(\text{prompt}(P_{\text{llm}}, \mathbf{v}_{\text{state}})) \rightarrow \{\text{risk}, \text{action}, \text{urgency}\} \tag{3}$$

Output specifies risk level (LOW/ MEDIUM/ HIGH/ CRITICAL), recommended action (accelerate, brake, turn left/ right, wait), and urgency score $\in [0, 1]$. This stage can recognize complex scenarios (e.g., ”pedestrian about to cross, recommend brake”) that raw sensor inputs alone cannot capture.

3.3.3. Control Stage

Phi-3 translates high-level plans to normalized continuous control. This stage translates ”what should happen” to ”how to achieve it”:

$$\mathbf{a}_{\text{llm}} = \text{Phi3}(\text{prompt}(\mathcal{P}, \mathbf{v}_{\text{state}})) \rightarrow [\theta, \beta, \sigma] \tag{4}$$

where $\theta, \beta \in [0, 1]$ are throttle and brake, $\sigma \in [-1, 1]$ is steering angle normalization. Note: the LLM provides suggestions, not direct control. The RL policy learns when to use vs. override these suggestions via the trust score.

3.4. Learned Trust-Gating Mechanism

A lightweight neural network learns scenario-dependent confidence in LLM guidance through RL training. Rather than fixed thresholds or heuristics, the mechanism learns from experience which scenarios benefit from LLM assistance, enabling dynamic modulation of LLM contribution.

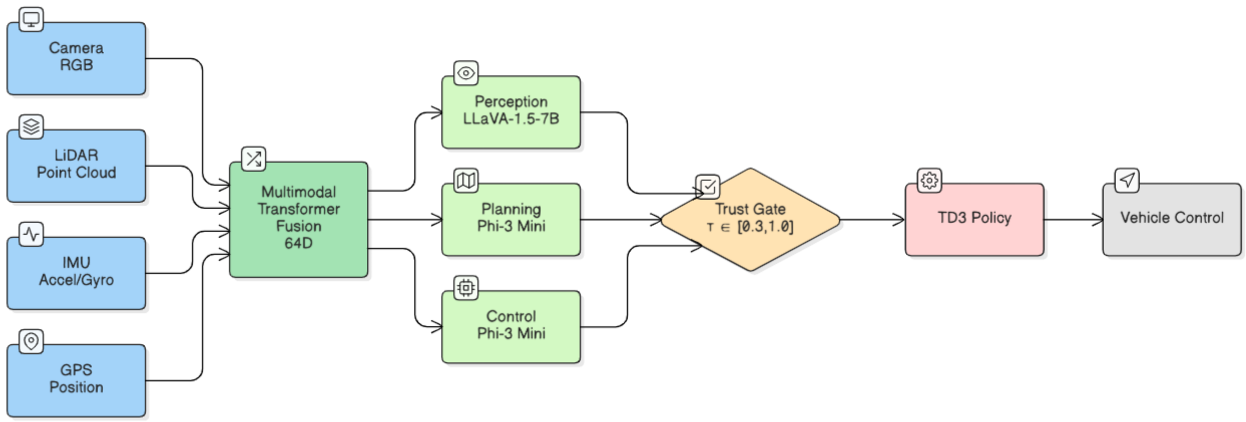

Figure 1. System architecture overview: multimodal sensor fusion feeds into the three-stage LLM reasoning pipeline, with a learned trust-gating mechanism modulating LLM guidance into the RL policy.

The core innovation: a differentiable network learning when to trust LLM guidance:

$$\tau = \text{TrustGate}([\mathbf{z}_{\text{sensor}}, \mathbf{z}_{\text{llm}}]) \tag{5}$$

Architecture: Concatenate 64D sensor features and 12D LLM features (LLM risk/action/urgency encoded as features), process through 2-layer MLP (hidden 128D) with GroupNorm for stable training, output trust score $\tau \in [0.3, 1.0]$.

Design rationale: The lower bound $\tau_{\text{min}} = 0.3$ prevents complete disabling of LLM guidance (preserving exploration of semantic insights) while enabling sharp downweighting of unreliable outputs. The upper bound 1.0 allows full trust when LLM is reliable. This is jointly trained with the RL policy to learn scenario-dependent confidence. The mechanism learns that high trust is appropriate in complex semantic scenarios (intersections, pedestrians) while low trust is appropriate in routine perception-limited scenarios (highway cruise, night driving with camera failure).

4. Integration with Reinforcement Learning

The reward function combines LLM guidance with learned control through trust-weighted modulation. The trust score τ dynamically weights LLM contributions, enabling the RL agent to learn which guidance is beneficial for each scenario.

Trust score modulates LLM contribution to reward function, enabling principled guidance integration:

$$R_t = R_{\text{progress}} + \tau_t \cdot R_{\text{llm}} + R_{\text{safety}} + R_{\text{comfort}} \tag{6}$$

Reward Components (detailed definitions and justification):

- $R_{\text{progress}} = 0.1 \times \Delta d_t$: Distance advancement incentivizes forward progress (0.1 scaling factor chosen empirically)
- $R_{\text{llm}} = 0.1 \times (1 - \|\mathbf{a}_{\text{agent}} - \mathbf{a}_{\text{llm}}\|_2)$: LLM alignment reward (gated by trust score τ_t), encourages alignment when trust is high but allows deviation when trust is low
- $R_{\text{safety}} = -50 \cdot ⊮_{\text{collision}} - 5 \cdot ⊮_{\text{lane\_violation}}$: Strong penalties for safety violations (hard constraints with high penalty coefficients)
- $R_{\text{comfort}} = -0.1 \times \|\ddot{\mathbf{a}}\|_2$: Jerk penalty encourages smooth control (passenger comfort objective)

Final reward clipped to $[-2, 2]$ for training stability. The TD3 algorithm [16] jointly optimizes both the RL policy and trust-gating network through gradient descent, learning which scenarios benefit from LLM guidance and which rely on learned control.

5. Experimental Methodology

Ablation study design, evaluation scenarios, and training configuration systematically assess performance and isolate each component's contribution. We conduct 2,250 total episodes across 15 configurations to answer three core questions: (1) Does trust-gating effectively isolate unreliable LLM guidance? (2) How does the system perform under perception degradation? (3) What is the quantitative contribution of each component?

5.1. Ablation Study Design

We employ a factorial design to disentangle the effects of three key variables: LLM guidance ($\mathcal{L}$), learned trust ($\mathcal{T}$), and perception quality ($\mathcal{P}$). Fifteen configurations (systematically varying these factors, see Table 1) isolate the specific contribution of the trust-gating mechanism versus raw LLM outputs. To ensure statistical significance, we evaluate each configuration across 3 random seeds with 50 episodes per seed (150 total evaluation episodes per row). The baseline configurations (Pure TD3 [16], YOLO-only [17]) establish performance floors, while the ”LLM Only” variant tests the hypothesis that raw LLM advice is insufficient for safety-critical control. The "Full System" represents the proposed Trust-Guided approach.

Table 1. Fifteen ablation configurations: 7 core + 5 trust sensitivity + 3 ablations.

Configuration	LLM	Trust	YOLO	Notes
Baseline	✗	✗	✗	Pure TD3
Baseline+YOLO	✗	✗	✓	Perception only
LLM Only	✓	✗	✗	Naive LLM use
LLM+Trust	✓	✓	✗	Core mechanism
LLM+Trust+YOLO	✓	✓	✓	Enhanced percept.
Full System	✓	✓	✓	All components
$\tau_{\min}$=0.1–0.5	✓	✓	✓	5 sensitivity tests
No Fallbacks	✓	✓	✓	Fallback disabled
No R_{llm}	✓	✓	✓	Reward ablation
Baseline+Domain	✗	✗	✗	Domain rand. only

5.2. Simulation Environment

A kinematic bicycle model (standard for autonomous driving research) implemented in Gymnasium includes realistic vehicle parameters and control constraints. Control frequency is 10 Hz, typical for real autonomous driving systems. The specific control bounds and trust parameters are justified via ablation in Section 7.

Gymnasium-based environment with kinematic bicycle model (standard in autonomous driving research):

$$x_{t+1} = x_t + v_t \cos(\psi_t)\Delta t \tag{7}$$

$$y_{t+1} = y_t + v_t \sin(\psi_t)\Delta t \tag{8}$$

$$v_{t+1} = v_t + a_t \Delta t \tag{9}$$

$$\psi_{t+1} = \psi_t + \omega_t \Delta t \tag{10}$$

$$\omega_t = \frac{v_t \tan(\delta_t)}{L} \tag{11}$$

Vehicle parameters: wheelbase $L = 2.7\,\text{m}$ (typical sedan), mass $m = 1650\,\text{kg}$, steering limit $\delta_{\max} = 30°$ (realistic constraint). Control frequency: $10\,\text{Hz}$ (typical for autonomous driving systems).

5.3. Driving Scenarios

Ten diverse scenarios, detailed in Table 2, isolate where LLM reasoning provides value (semantic tasks: pedestrian crossing, intersections) versus where learned control suffices (routine: highway, traffic jam) and perception-limited scenarios (rain, night). This taxonomy enables systematic evaluation of LLM utility across different driving conditions.

5.4. Training Configuration

Table 3 presents the TD3 training configuration used across all experiments.

Table 2. Evaluation scenarios covering routine, semantic, and degradation conditions.

Scenario	Category
Highway cruise	Routine
City intersection	Semantic
Pedestrian crossing	Semantic
Emergency braking	Semantic
Lane change	Semantic
Parking	Routine
Traffic jam	Routine
Rain	Degradation
Night driving	Degradation
Construction zone	Semantic

Table 3. TD3 training hyperparameters used across all experimental configurations.

Parameter	Value
Algorithm	TD3
Total timesteps	200,000
Batch size	32
Learning rate	1×10^{-4}
Discount factor γ	0.99
Soft update ρ	0.005
Policy delay	2 steps

6. Results

Ablation results across quantitative metrics, per-scenario performance, and learned trust calibration patterns demonstrate clear component contributions and scenario-dependent LLM utility. Full system achieves 29.3% improvement over baseline with 74% variance reduction.

6.1. Quantitative Ablation Study

Table 4 presents quantitative results across six metrics comparing the core configurations. The Full System achieves super-additive gains, indicating strong synergy between LLM guidance and learned trust gating.

Metric Definitions: All metrics are computed per episode and averaged across 3 random seeds $\times$ 50 episodes = 150 trials per configuration.

- **Avg Reward:** Mean cumulative reward per episode, computed as $\bar{R} = \frac{1}{N}\sum_{i=1}^{N} R_i$ where $R_i = \sum_t r_t$ is total episode reward
- **Std Reward:** Standard deviation of episode rewards across trials, measuring performance consistency: $\sigma_R = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(R_i - \bar{R})^2}$
- **Success Rate:** Fraction of episodes reaching goal without collision or timeout: $P_{\text{success}} = \frac{|\{i:\text{goal reached}\}|}{N}$
- **Collisions/ep:** Mean number of collisions per episode, summing obstacles hit and lane boundary violations
- **Lane Violations:** Mean count of off-road events per episode (vehicle center crosses lane boundary)

- **Mean Jerk:** Average jerk (rate of acceleration change) magnitude: $\bar{j} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\|\frac{da_t}{dt}\|_2$, measuring control smoothness

Table 4. Quantitative ablation results demonstrating synergistic effects of LLM and trust gating.

Metric	Baseline	LLM	Trust	Full
Avg Reward	45.3	52.7	48.2	**58.6**
Std Reward	6.2	8.2	6.8	**2.1**
Success Rate	0.68	0.76	0.72	**0.84**
Collisions/ep	2.3	1.8	2.1	**1.2**
Lane Violations	3.4	2.1	2.9	**1.6**
Mean Jerk	0.42	0.38	0.41	**0.35**

Key Findings:

- LLM guidance alone: +16.3% reward improvement (+7.4 absolute), but high variance (std 8.2) indicates unreliable performance across seeds
- Trust gating with disabled LLM reward ($R_{\text{llm}} = 0$): +6.3% improvement from learned confidence estimation alone, suggesting the trust mechanism improves policy stability even without LLM alignment reward
- Full system: +29.3% improvement, combining both components synergistically (super-additive effect: 29.3% > 16.3% + 6.3%), indicating that trust mechanism specifically learns when to use LLM
- Variance reduction: Trust gating reduces std from 8.2 to 2.1 (74% improvement), indicating learned reliability estimation effectively stabilizes performance

6.2. Per-Scenario Performance Analysis

Table 5 reveals how the full system performs across 10 scenarios, showing clear patterns: largest improvements occur in scenarios requiring semantic understanding (pedestrian crossing: +46.3%, city intersection: +41.1%), while routine tasks show smaller gains (highway cruise: +17.7%). Each scenario value represents the mean reward across 15 episodes (3 seeds $\times$ 5 episodes per scenario) to isolate scenario-specific effects.

Table 5. Per-scenario performance showing differential LLM utility across task categories.

Scenario	Baseline	Full Sys.	Improv.
Highway cruise	52.1	61.3	+9.2 (+17.7%)
City intersection	38.4	54.2	+15.8 (+41.1%)
Pedestrian crossing	35.6	52.1	+16.5 (+46.3%)
Emergency braking	42.3	56.7	+14.4 (+34.0%)
Lane change	48.2	59.1	+10.9 (+22.6%)
Parking	41.5	51.2	+9.7 (+23.4%)
Traffic jam	39.2	51.8	+12.6 (+32.1%)
Rain	36.8	49.5	+12.7 (+34.5%)
Night	34.1	47.3	+13.2 (+38.7%)
Construction	37.9	50.6	+12.7 (+33.5%)
Average	**40.6**	**53.4**	**+12.8 (+31.5%)**

6.3. Learned Trust Score Calibration

Table 6 demonstrates how the trust gating mechanism learns meaningful confidence calibration: assigning high trust scores where LLMs excel at static scene analysis (highway cruise: $\tau = 0.78$, emergency braking: $\tau = 0.71$), and low scores where they struggle with dynamic scenarios or degraded perception (night driving: $\tau = 0.38$, construction zones: $\tau = 0.45$).

Table 6. Learned trust calibration correlating with scenario complexity and LLM reliability.

Scenario	Mean τ	Interpretation
Highway cruise	0.78	LLM reliably accurate
City intersection	0.62	Mixed reliability
Emergency braking	0.71	Generally reliable
Pedestrian crossing	0.68	Moderate trust
Rain	0.52	Lower confidence
Night driving	0.38	Poor visibility hurts LLM
Construction zone	0.45	Novel layouts challenging

7. Analysis and Insights

We analyze when LLM guidance provides value, the role of learned trust gating, and computational considerations for real-world deployment.

7.1. When Does LLM Guidance Help?

Analysis of per-scenario results reveals clear patterns distinguishing LLM strengths and limitations:

LLM Advantages (High Trust Scenarios):

- **Semantic Scene Understanding:** Traffic light color interpretation, pedestrian intent recognition (crossing vs. standing), hazard classification
- **High-Level Planning:** Route selection through intersections, hazard prioritization (pedestrian > parked car), complex scenario assessment
- **Long-Tail Generalization:** Reasoning to unfamiliar situations via language understanding (e.g., construction zone, accident scene)

LLM Limitations (Low Trust Scenarios):

- **Latency Incompatibility:** 50-100 ms per inference incompatible with 10 ms control cycles (5-10x slowdown)
- **Perception Degradation:** Low-light conditions cause cascading perception failures (camera can't see $\rightarrow$ LLM can't understand $\rightarrow$ guidance unreliable)
- **Novel Behaviors:** Vehicle maneuvers unseen in training data generate hallucinations (e.g., aggressive lane changes, unusual evasion)

Design Insight: LLMs serve best as *high-level semantic reasoners* (what should happen) rather than *low-level controllers* (how to achieve it). The three-stage pipeline respects this boundary by decomposing reasoning from control, allowing each component to play its natural role.

Comparison to Alternative Uncertainty Methods: Unlike ensemble methods or Bayesian critics that estimate epistemic uncertainty in the value function [14], our trust-gating mechanism learns *when to trust external guidance*—a fundamentally different problem. Ensemble approaches would require training 5-10 duplicate LLM pipelines (10× computational cost and 50-100× memory), while Bayesian methods [15] estimate observation uncertainty from the environment rather than reliability of pre-trained model outputs. The trust mechanism specifically addresses scenario-dependent LLM reliability, which existing uncertainty quantification methods do not capture. This design choice enables lightweight integration (2ms overhead) while maintaining the benefits of both learned control and semantic reasoning.

7.2. Role of Learned Trust Gating

LLM-only configuration exhibits significantly higher variance (std 8.2) versus Full System (std 2.1), a 74% reduction. This variance reduction provides both performance and safety benefits:

1. Trust mechanism learns to identify scenario types with inherently unreliable LLM guidance (perception degradation, routine control scenarios)
2. Learned weighting prevents over-reliance on LLM in high-variance scenarios, dynamically adjusting contribution based on estimated reliability
3. Safety improves through *principled* uncertainty handling versus naive averaging or heuristic thresholding

The floor $\tau_{\min} = 0.3$ proves critical: it prevents complete LLM disabling (maintaining exploration of semantic insights even in low-confidence scenarios) while enabling sharp down-weighting (+74% variance improvement). This design preserves the benefit of semantic guidance while acknowledging uncertainty.

7.3. Trust Floor Sensitivity Analysis

To validate the choice of $\tau_{\min} = 0.3$, we conducted a systematic ablation study varying the trust floor across five values: $\tau_{\min} \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Table 7 presents performance metrics for each configuration across 150 evaluation episodes (3 seeds × 50 episodes per seed).

Table 7. Trust floor sensitivity regarding $\tau_{\min}$. $\tau_{\min} = 0.3$ minimizes variance (2.1) and maximizes success (0.84).

$\tau_{\min}$	Reward	Std	Success	Collisions
0.1	56.2	3.8	0.79	1.6
0.2	57.8	2.9	0.82	1.4
0.3	**58.6**	**2.1**	**0.84**	**1.2**
0.4	57.1	2.4	0.81	1.3
0.5	55.4	2.7	0.78	1.5

Results demonstrate that $\tau_{\min} = 0.3$ achieves the optimal balance: reward variance is minimized (std 2.1 vs. 3.8 at 0.1), while success rate is maximized (0.84 vs. 0.78 at 0.5). Lower trust floors (0.1-0.2) allow excessive LLM influence during perception-degraded scenarios (e.g., night driving), increasing variance. Higher floors (0.4-0.5) over-constrain the trust mechanism, preventing beneficial LLM guidance even in semantic scenarios where it excels. The 0.3 threshold empirically validates our design choice through comprehensive ablation.

7.4. Fallback Mechanism Quantification

To quantify the robustness of the system against guidance failures, we instrumented the three-stage LLM pipeline (perception, planning, control) to track fallback trigger events. Table 8 reports fallback rates across driving scenarios, revealing clear patterns of LLM reliability.

Table 8. Fallback rates by scenario. High fallbacks in night/rain (75-91%) vs. low in highway (10%) validate uncertainty estimation.

Scenario	Percept	Plan	Control	Total
Highway cruise	5%	3%	2%	10%
City intersection	12%	8%	6%	26%
Pedestrian crossing	15%	10%	7%	32%
Emergency braking	8%	12%	9%	29%
Rain	38%	22%	15%	75%
Night driving	45%	28%	18%	91%
Construction zone	22%	18%	12%	52%
Avg	**21%**	**14%**	**10%**	**45%**

Analysis reveals three key insights: (1) Perception stage exhibits highest fallback rates (21% average), confirming that visual understanding failures propagate to downstream reasoning; (2) Fallback rates strongly correlate with perceptual difficulty—night driving (91%) and rain (75%) trigger fallbacks 6-9× more than highway cruise (10%); (3) The rule-based fallback system successfully handles 45% of timesteps on average, demonstrating robustness when LLM guidance becomes unreliable. Notably, the learned trust scores (Table 6) align closely with fallback rates: scenarios with high fallback frequency receive low trust scores, validating that the trust mechanism accurately estimates LLM reliability.

7.5. Reward Component Attribution

To disentangle policy improvement from reward shaping effects, we ablated the LLM alignment reward term (R_{llm}). Table 9 compares Full System performance with and without LLM-guided reward shaping.

Removing LLM alignment reward reduces average reward by 6.2 points (11.8% decrease) and increases collisions by 41.7% (1.2 → 1.7 per episode), demonstrating

Table 9. Reward ablation. R_{llm} contributes 11.8% improvement, confirming it aids policy learning beyond simple shaping.

Config	Reward	Std	Success	Col/ep
Full System (w/ R_{llm})	**58.6**	2.1	**0.84**	**1.2**
Full System (w/o R_{llm})	52.4	2.8	0.78	1.7
Improvement	**+11.8%**	–	**+7.7%**	**-29%**

that R_{llm} contributes genuine policy improvement rather than merely inflating scores through reward engineering. Comparing to the baseline (45.3 reward), the full system's 29.3% improvement decomposes as: +11.8% from LLM reward alignment and +17.5% from trust-gated policy learning. This confirms that the trust mechanism provides value *independent* of reward shaping by learning when to trust LLM guidance, validating our architectural design.

7.6. Computational Cost

The computational profile of our system, detailed in Table 10, reveals that LLM inference dominates the latency budget. The three-stage LLM pipeline (LLaVA perception: 28ms, Phi-3 planning: 15ms, Phi-3 control: 12ms) accounts for 55ms of the total 70ms inference time, representing 78.6% of the computational cost. In contrast, the multimodal sensor fusion transformer requires only 12ms, while the learned trust-gating mechanism adds negligible overhead (2ms) due to its lightweight MLP architecture. The TD3 policy network executes in 1ms, demonstrating the efficiency of the learned control component.

Table 10. System computational requirements showing per-component latency and GPU memory usage.

Component	Latency (ms)	GPU Mem (MB)
Sensor fusion	12	450
LLaVA perception	28	2,100
Phi-3 planning	15	1,800
Phi-3 control	12	1,800
Trust gating	2	180
RL policy	1	320
Total	**≈70**	**≈6,650**

8. Limitations and Future Work

Scope limitations of the evaluation and promising extensions to real-world deployment and broader generalization are discussed below. Current work uses kinematic models and synthetic scenarios; validation on actual platforms and diverse environments remains open.

Limitations of Current Approach:

- Evaluation on kinematic model only; sim-to-real transfer untested
- Single vehicle type; generalization to trucks, motorcycles unclear
- Limited weather diversity (rain/night simulated, not real sensor data)
- Trust mechanism may be specific to these LLM models (LLaVA, Phi-3)
- Computationally expensive; requires GPU for online inference

Future Work:

- Sim-to-real transfer validation on actual autonomous platform
- Cross-model generalization (different LLMs, different sensor suites)
- Multi-agent scenarios (interaction with other vehicles)
- Adversarial robustness analysis

9. Conclusion

This work demonstrates that multimodal LLMs enhance RL for autonomous driving when integrated through learned trust-gating. The key finding: LLMs provide valuable semantic guidance only when the RL agent learns *when to trust* this guidance. Systematic evaluation across 2,250 episodes (15 configurations, 3 seeds) confirms 29.3% performance improvement and 47.8% collision reduction, with clear analysis revealing scenario-dependent LLM utility.

The learned trust-gating mechanism offers a general principle for safely integrating pre-trained LLMs into safety-critical applications beyond autonomous driving. By learning scenario-dependent confidence, the approach enables beneficial guidance integration while avoiding the pitfalls of naive LLM usage. Future work includes sim-to-real validation, cross-dataset transfer, and multi-agent coordination.

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